

Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with a passkey: Face ID, Touch ID or a password manager.

Live (Stellar TESTNET)

kumbara.sembol.xyz

Repository

github.com/keyboard01/kumbara

Deck date

6 September 2026

Lira savings lose value; the dollar route has a wall in front of it.

Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.¹ Dollar-denominated saving is how ordinary households defend themselves.

The crypto route exists but sheds users at every step

USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

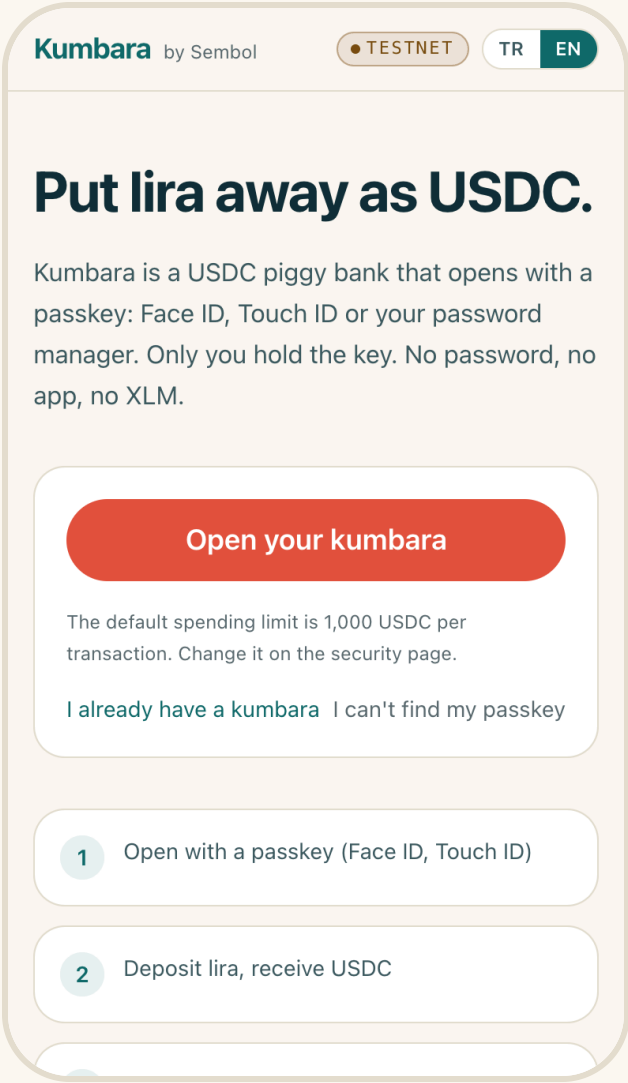
The people who need hard-currency savings most are served least

A bank-transfer habit and a phone with a passkey (Face ID or Touch ID) are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

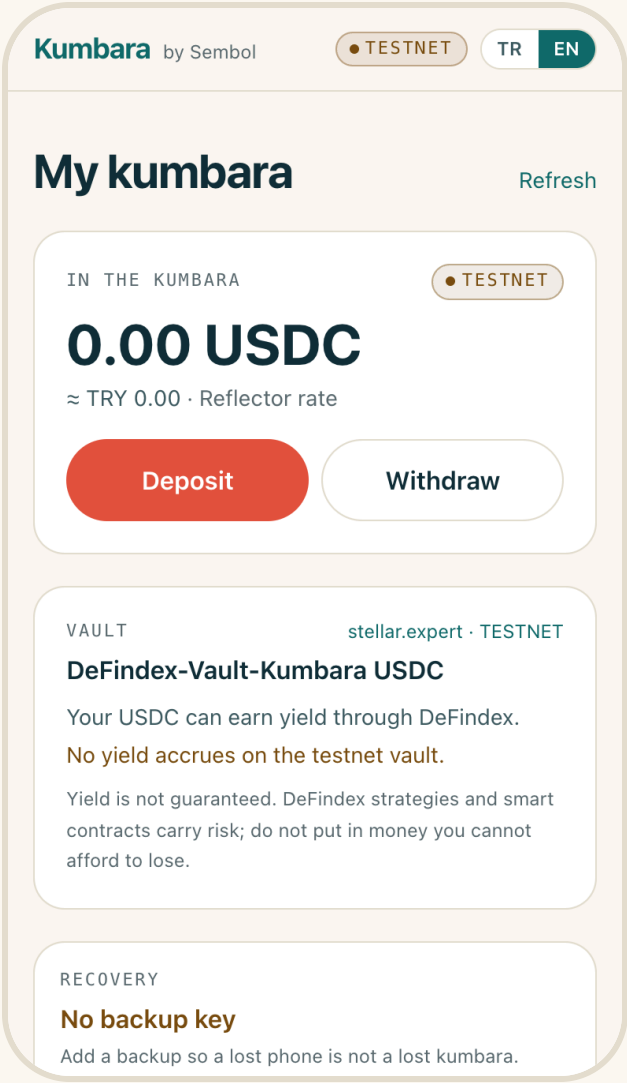
Türkiye is market one. The product is built for a class of markets: high-inflation economies with a bank-transfer habit and an existing Stellar anchor. The anchor is configuration, not code, so the same app serves the next such market.

¹ TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% against August 2025 (monthly +1.84%). Listed on tuik.gov.tr ("%31,51 (yıl)") with the bulletin at veriportali.tuik.gov.tr/tr/press/58290; verified on tuik.gov.tr on 6 September 2026.

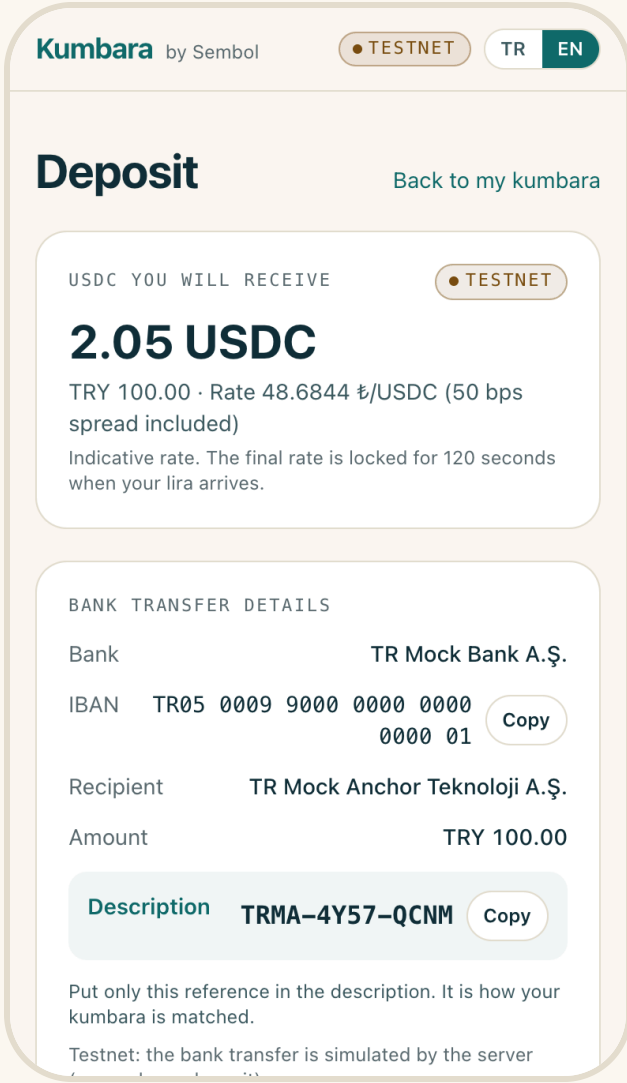
Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



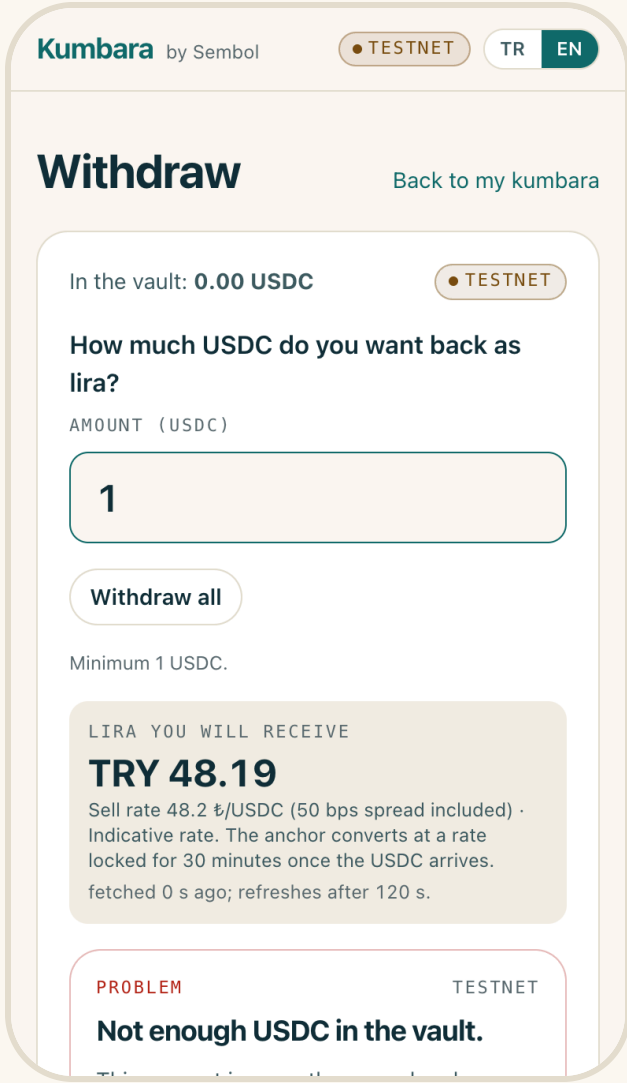
Onboard. One button; a passkey (Face ID, Touch ID, password manager) creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



Savings. USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



Deposit. IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



Withdraw. A sell quote in lira; two passkey approvals; the anchor pays the visitor's own IBAN.

Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

- 1

The visitor opens an OpenZeppelin smart account with a passkey (Face ID, Touch ID); the relay pays the fees.
- 2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.
- 3

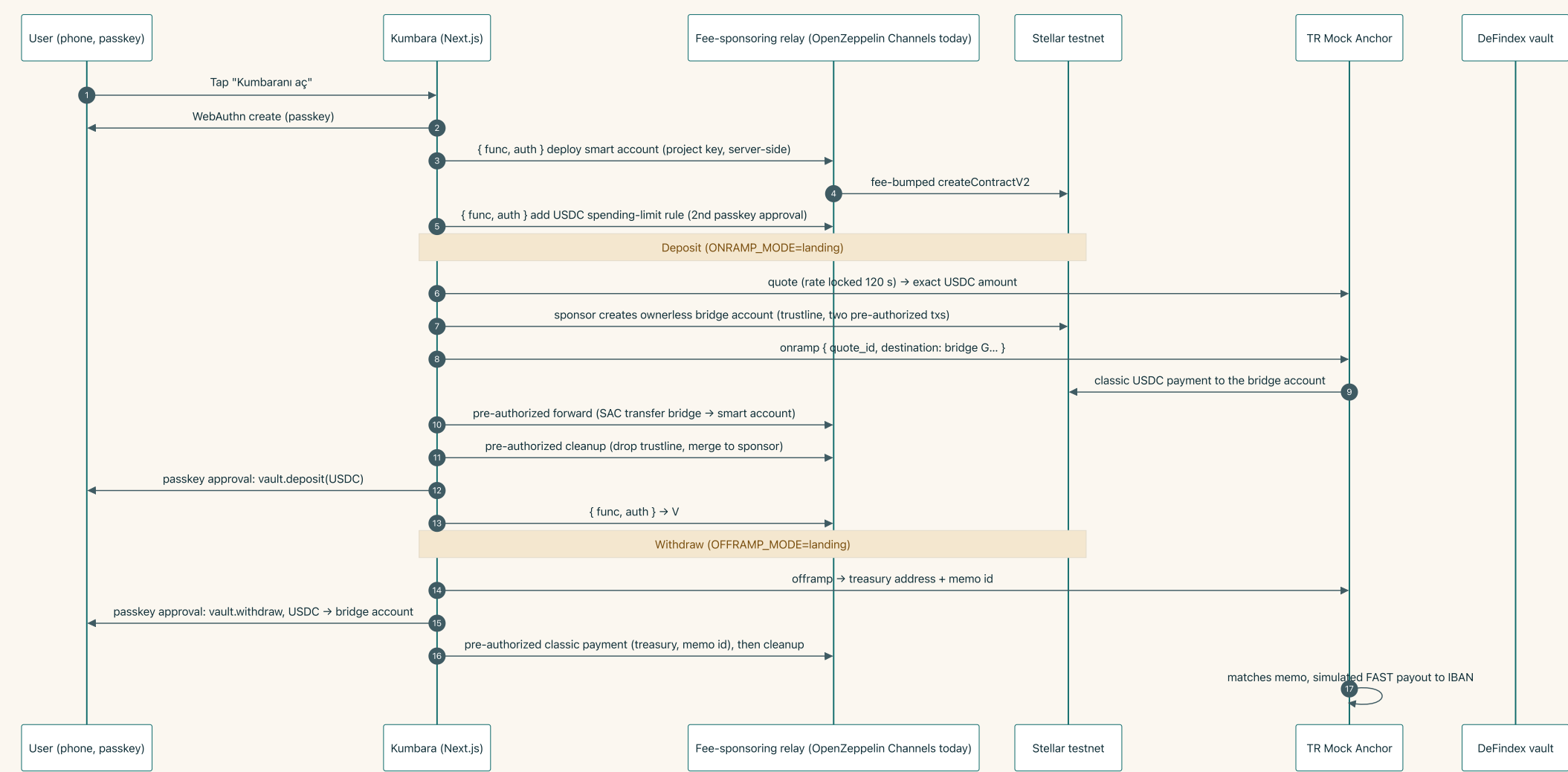
The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).
- 4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.
- 5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.
- 6

One passkey approval puts the USDC into the DeFindex vault (no yield on testnet).
- 7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.



Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

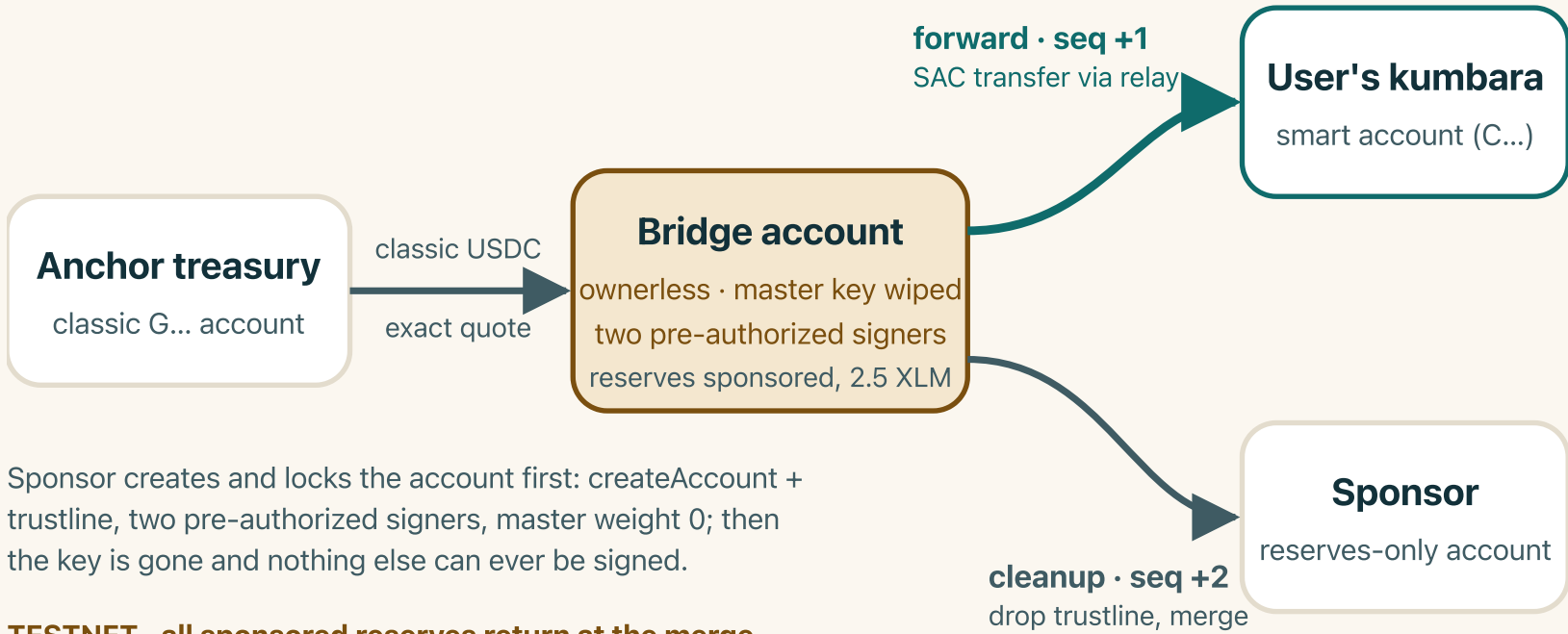
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

TESTNET · all sponsored reserves return at the merge

EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...)	TRANSACTION
Anchor pays the bridge account (classic payment)	778215b3...e3c3a8f1
Pre-authorized forward: USDC lands in the smart account	f580e030...f22c9eab
Pre-authorized cleanup: trustline removed, account merged	6223c61c...7416291d

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

BUILDING BLOCK	WHAT IT DOES IN THIS APP	ADDRESS (TESTNET, STELLAR.EXPERT)
OpenZeppelin smart accounts + spending-limit policy	Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit.	WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G
DeFindex vault + hodl strategy	Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.	vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN
TR Mock Anchor · SEP-1/6/38 + Partner API	The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.	USDC SAC CBIELTK6...XQDAMA
Fee-sponsoring relay (OpenZeppelin Channels today)	Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.	channels.openzeppelin.com/testnet
Reflector*	The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).	CBKGPWGK...4DXMCJZC (mainnet oracle, read only)*

Upstream: "Fix wallet creation crash on default Next.js apps" was merged into [stellar/smart-account-kit](#) on 19 August 2026 and shipped in 0.6.2 · [github.com/stellar/smart-account-kit/pull/4](#)

Open-source plan: the repository is MIT-licensed. Kumbara deploys no proprietary contracts: the smart account, WebAuthn verifier and spending-limit policy are OpenZeppelin's audited contracts, the vault and strategy are DeFindex's; the bridge-account logic ([lib/landing](#)) is MIT in the repo with its tests, and the repository goes public at award time.

* Reflector is a mainnet price oracle read by simulation only: Kumbara sends no mainnet transaction and makes no mainnet claim; every balance here is Stellar TESTNET and the anchor's quote is the fallback rate.

Everything on this slide is public, linkable and re-runnable.

Kumbara by Sembol

TESTNETTREN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event startAll time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY inTESTNET

TRY 0.00

TRY outTESTNET

26.79

USDC

USDC in the vault (live)

TESTNET

kumbara.sembol.xyz/stats: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 ·
100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

E2E · passing · latest production run 6
Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

ON-CHAIN EVIDENCE (STELLAR TESTNET)

TRANSACTION

Vault deployed through the DeFindex factory with the hodl strategy attached

dbf0d5b7...6faa8f96

Deposit with invest=true moved USDC into the strategy (invest)

b615e7dc...76b7f981

Withdrawal unwound the shares from the strategy (divest)

466f48d2...07a9f5ca

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on kumbara.sembol.xyz, 2:39 with a caption per step.

Non-custodial software in front of a licensed anchor. Built for any market that has one; Türkiye is market one.

- **Kumbara never holds lira, USDC or keys.** Every transfer is authorized by the visitor's own passkey on their own smart account; the bridge accounts have no owner.
- **The licensed anchor is the regulated party:** it holds the fiat, performs KYC and pays out only to the customer's own named bank account.
- **No payments feature:** no send, no P2P, no merchant flow. Deposit into a vault and withdraw back to your own account, nothing else.
- **Local AML rules shape the UX** wherever it launches: named accounts in and out, an on-chain per-transaction cap, anchor-side limits, no cash or third-party paths.
- **The regulatory gate is the anchor's own position:** the selected anchor's written compliance position on a non-custodial front end, obtained during the Tranche 1 go/no-go. No paid counsel; no entity is needed to receive the award.

ANY LICENSED STELLAR ANCHOR · THE ANCHOR IS A CONFIG CHOICE

Issuer and endpoints come from the anchor's stellar.toml, deposits and withdrawals run on SEP rails, and the bridge accounts pay any classic account, so they are anchor-agnostic. The vault and wallet layer never change. Per market: copy, currency display and the rate oracle.

Market class: high-inflation economies with a bank-transfer habit and an existing Stellar anchor. **Türkiye first, by fit:** 31.5% annual inflation (TÜİK, August 2026), a transfer habit everyone already has, and the TRY sandbox rails this deck demonstrates.

GO / NO-GO IN TRANCHE 1 · FALLBACK BUILT IN

Tranche 1 ranks the on/off-ramps on the SCF Integration List by market: licensed status, bank-transfer rails to the customer's own account, a written non-custodial position, and whether a counterparty entity is required (a shortlist criterion, not a cost). Türkiye stays first only with a licensed TRY anchor that passes; otherwise the pilot launches in the top-ranked market and the app does not change.

The same shortlist is the market-two shortlist. A second market is the natural follow-on award once the metric is hit; it is not promised inside this one.

One builder, already shipping in the ecosystem.

Ahmed

Founder · builds and pitches

Instaward grantee (Stellar Türkiye). Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on, and contributor to [stellar/smart-account-kit](#). Designed and built Kumbara end to end: the app, the bridge accounts, the CI round trips and the evidence in this deck.

Site [ahmed.bet](#)

LinkedIn [linkedin.com/in/ahmed-zm](#)

GitHub [github.com/keyboard01](#)

SHIPPED UPSTREAM

stellar/smart-account-kit #4: "Fix wallet creation crash on default Next.js apps", merged 19 August 2026, in release 0.6.2.

PUBLISHED

@sembol/passkey-react on npm: passkey creation and signing, recovery, spending-limit policy UI. Every screen in this deck runs on it.

FUNDED

Instawards grantee. Kumbara is the consumer proof for that work, built for this submission.

Wallet layer and relay hardening first, then testnet-complete and secured, then mainnet in the selected pilot market.

TRANCHE 1 · DAYS 1–60 Wallet layer and relay hardening me 340 h · no hired lines pays 20% at completion (MVP)	- Single-transaction onboarding: spending-limit policy on the default rule at creation; the library's limit UI updated @sembol/passkey-react 0.4 published with projectId: relay project binding, config, docs, migration notes - Relay mainnet hardening of the fee-sponsoring path: per-project budgets, allowlists, refusal alerts, sponsor monitoring - Anchor shortlist and go/no-go: on/off-ramps ranked by market, written positions from the top two, pilot anchor chosen, runner-up named - Audit Bank application submitted with scope: policy install, bridge accounts, server path Shortlist input, unranked until this tranche (SCF Integration List, on/off-ramping): Koywe · alfredpay · Abroad · BlindPay · Mercuryo · Bridge. No TRY anchor is listed today; Türkiye stays market one only if a licensed TRY anchor passes the same criteria, otherwise the pilot launches in the top-ranked market and the app does not change. The runner-up is a shortlist output, not an integration in this award.
TRANCHE 2 · DAYS 61–120 Testnet-complete and secured me 340 h + hired 80 h pays 30% at completion (testnet)	- Selected anchor wired in via its sandbox or API in place of the mock anchor (SEP-6/24 and SEP-38 where offered) - Threat model in SDF's STRIDE template format: assets, threat table, mitigations - Monitoring plan (builders template) with core alerts live for vault, policy, sponsor and bridge accounts; dashboards hired, optional - Audit Bank audit of the policy install, bridge accounts and server path; findings remediated with regression tests; QA hired - Mainnet path behind a capped, presenter-only flag with real USDC; hosted and self-host deployment docs with scripted setup
TRANCHE 3 · DAYS 121–180 Mainnet launch and public pilot me 340 h + hired 80 h pays 40% at completion (mainnet)	- Mainnet launch in the pilot market by week 3: contracts registered, sponsor and relay budgets live, production rails - Public pilot cohort of 50–100 users; validation-focused user testing run by me; session ops and UX polish hired (1/3 exception) - Monitoring stack live on mainnet with an on-call runbook: rotation, escalation, incident templates - Metric window: 30-day NAV measurement, public dashboard, on-chain footprint registered - Pilot support process in operation: support channel, triage rules, response targets, fixes, weekly pilot reports

Boundary with the Sembol Instaward. Instaward Month 3 delivers the testnet relay. This award funds only what lies beyond it: the mainnet hardening of that fee-sponsoring path (per-project budgets, allowlists, refusal alerts, sponsor monitoring), the 0.4 library release and policy-based single-transaction onboarding. No hour appears in both.

BUDGET AND METRIC · \$107,300 · INTEGRATION TRACK LARGE BAND (\$100K–150K), AT ITS LOW END · EVERY DOLLAR IS HOURS

One full-time engineer plus hired capacity: 1,180 hours over six months, \$107,300.

TRANCHE	DAYS	ME H	HIRED H	ME USD	HIRED USD	SUBTOTAL
1 · Wallet layer and relay hardening	1–60	340	0	\$32,300	\$0	\$32,300
2 · Testnet-complete and secured	61–120	340	80	\$32,300	\$5,200	\$37,500
3 · Mainnet launch and public pilot	121–180	340	80	\$32,300	\$5,200	\$37,500
Total · six months	180	1,020	160	\$96,900	\$10,400	\$107,300

Paid in XLM per the handbook split: 10% on acceptance \$10,730 · 20% after Tranche 1 \$21,460 · 30% after Tranche 2 \$32,190 · 40% after Tranche 3 \$42,920

HIRED CAPACITY · 160 H AT \$65/H · PARALLEL WORK ONLY; NO TRANCHE GATE DEPENDS ON A HIRE

Monitoring dashboards and alert routing, optional (Tranche 2)	40 h	\$2,600
QA against the audit findings: independent re-test, regression suite (Tranche 2)	40 h	\$2,600
User-testing ops: structured sessions, interviews, write-ups (Tranche 3)	40 h	\$2,600
UX polish for the pilot: copy, edge states, accessibility fixes (Tranche 3)	40 h	\$2,600

If a hire does not land, the line is unbilled and the tranche still closes on my own lines: core alerts, my regression tests, the sessions I run, fixes from pilot support. The dashboards line is optional by design.

FINAL-TRANCHE METRIC COMMITMENT

Metric type	DeFindex vault NAV: USDC in Kumbara's mainnet vault position, from local-currency deposits in the pilot market
Threshold	average daily NAV of 2,500 USDC = 100 pilot users × 50 USDC average balance × 50% still holding through the window. The cohort is planned at 50–100 users; the threshold takes the top of that range
Window	the 30 days after the pilot launch, averaged over the window, not a snapshot
Footprint, registered at award time	vault, strategy and policy contracts; sponsor account (bridge accounts sponsored per CAP-33); app-created smart accounts

The panel ratifies or adjusts the threshold during panel review. Seed and E2E accounts are tagged out of every number; wash or gamed activity is excluded.

BUDGET HYGIENE

Only development, testing, monitoring, deployment and validation work is in the ask; nothing is self-funded and every line is hours × rate. No legal fees, no entity setup (none is needed to receive the award), no audit fee (Audit Bank), no marketing beyond the user-testing exception: validation lines are about 8% of the total against the one-third cap. Nothing waits on a third party signing, and nothing here is paid for by the Instaward.

Paid in XLM in tranches of 10% on acceptance, 20% at MVP, 30% at testnet and 40% at mainnet; "Project timeline should not exceed 6 months"; each tranche completion form within 90 calendar days of the previous payment: SCF Handbook, Build Award, "Budget & Deliverable Guidelines" (stellar.gitbook.io/scf-handbook/scf-awards/build-award/budget-and-deliverable-guidelines). Rates: \$95/h for a senior full-stack engineer with Soroban and passkey work shipped upstream (stellar/smart-account-kit PR #4; @sembol/passkey-react on npm), full-time at 40 h/week, 340 h per 60-day tranche; hired capacity at \$65/h for parallel work only.

20 lines, 1,180 hours, \$107,300. Each line names how reviewers verify it.

DELIVERABLE	HOW REVIEWERS VERIFY	WHO · HOURS	USD
TRANCHE 1 · WALLET LAYER AND RELAY HARDENING · DAYS 1–60		ME 340 H	\$32,300
Single-transaction onboarding: the spending-limit policy installed on the default rule at account creation; the library's limit UI updated	one tx hash per new kumbara on testnet; onboarding timing on /stats	me · 90 h	\$8,550
@sembol/passkey-react 0.4 published with projectId: relay project binding, config, docs, migration notes; Kumbara moved onto it	npm 0.4.x; changelog; Kumbara lockfile	me · 70 h	\$6,650
Relay mainnet hardening of the fee-sponsoring path: per-project budgets, allowlists, refusal alerts, sponsor monitoring	config in repo; a refused call's log and tx; an alert fired; sponsor monitor	me · 100 h	\$9,500
Anchor shortlist and go/no-go: Integration List on/off-ramps ranked by market; written positions from the top two; pilot anchor and market selected; runner-up named as the fallback	ranked shortlist, decision memo, written positions (repo)	me · 60 h	\$5,700
Audit Bank application with scope: policy install, bridge accounts, server path	submission confirmation; scope document	me · 20 h	\$1,900
TRANCHE 2 · TESTNET-COMPLETE AND SECURED · DAYS 61–120		ME 340 H + HIRED 80 H	\$37,500
Selected anchor wired in via its sandbox or API in place of the mock anchor (SEP-6/24, SEP-38 where offered; adapter otherwise)	sandbox round trip green in CI	me · 100 h	\$9,500
Threat model in SDF's STRIDE template format (assets, threat table, mitigations)	docs/security/threat-model.md	me · 50 h	\$4,750
Monitoring plan in the builders template; core alerts implemented for the vault, policy, sponsor and bridge accounts	monitoring-plan.md; a live alert on testnet	me · 50 h	\$4,750
↳ Monitoring dashboards and alert routing, optional: boards per contract and account, paging rules	dashboard URL; a routed alert	hired · 40 h	\$2,600
Audit Bank audit of the policy install, bridge accounts and server path: scoping, support, findings remediated with regression tests (fee: Audit Bank)	audit report; fix commits; auditor re-check	me · 80 h	\$7,600
↳ QA against the audit findings: independent re-test, regression suite extended	QA report; CI suite	hired · 40 h	\$2,600
Mainnet path behind a capped, presenter-only flag: real USDC, enforced budgets	mainnet tx hashes from the capped path	me · 30 h	\$2,850
Hosted (Vercel + Turso) and self-host (Fly + SQLite) deployment docs with scripted setup; CI against the anchor sandbox	docs URLs; a fresh deploy from the docs; CI run	me · 30 h	\$2,850
TRANCHE 3 · MAINNET LAUNCH AND PUBLIC PILOT · DAYS 121–180		ME 340 H + HIRED 80 H	\$37,500
Mainnet launch in the selected pilot market by week 3: contracts registered, sponsor account and relay budgets live, production rails	mainnet contract IDs on stellar.expert	me · 90 h	\$8,550
Public pilot cohort of 50–100 users; validation-focused user testing, run by me (within the 1/3 exception)	/stats on mainnet; tx hashes; cohort report	me · 60 h	\$5,700
↳ User-testing ops: structured sessions, interviews, write-ups	session log; cohort report	hired · 40 h	\$2,600
↳ UX polish for the pilot: copy, edge states, accessibility fixes from the sessions	fix commits; Lighthouse accessibility score	hired · 40 h	\$2,600
Monitoring stack live on mainnet with an on-call runbook: rotation, escalation, incident templates	runbook in repo; a mainnet alert and its ticket	me · 50 h	\$4,750
Metric window: 30-day NAV measurement, public dashboard, on-chain footprint registered	NAV dashboard URL; registered IDs	me · 40 h	\$3,800
Pilot support process in operation: support channel, triage rules, response targets, fixes, weekly pilot reports	issue log; weekly reports in the repo	me · 100 h	\$9,500
Total · Integration Track Large band (\$100k–150k), at its low end · me 1,020 h (\$96,900) + hired 160 h (\$10,400)		1,180 h	\$107,300